SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Solution Requirements

Date	11 July 2026
Team ID	
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users access EduGenie without mandatory login. Optional session tokens via Google OAuth 2.0 may be used to save learning history securely.	High
2	Authorization Levels	Two roles: Student (can ask questions, generate quizzes, get recommendations) and Admin (can manage content, view usage stats, configure AI models).	High
3	External Interfaces	Google Gemini API for AI inference; SQLite/PostgreSQL for data storage; HTML+CSS frontend communicates with FastAPI backend via REST endpoints.	High
4	Transaction Processing	All user queries are processed as API transactions — input is sent to Gemini AI, response is returned and optionally stored in the user session database.	High
5	Reporting	System generates learning activity summaries, quiz score reports, and topic-wise progress dashboards viewable by the user on the frontend.	Medium
6	Business Rules	Quiz generation is limited to educational topics only. Learning path recommendations are personalised based on topic input and difficulty level selected.	Medium
7	Compliance to Laws / Regulations	User data handled per GDPR guidelines. No personal data is shared with third parties. API keys and credentials are stored securely using environment variables.	High
8	Other	System must support multi-subject queries (Science, Maths, History, Programming, etc.) and provide responses in simple, student-friendly language.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	EduGenie must respond to user queries within acceptable time limits. FastAPI's async architecture ensures non-blocking request handling.	API response in < 3 seconds; page load < 2 seconds on standard broadband.
2	Scalability	The system must handle multiple concurrent users without degradation. Stateless FastAPI backend allows horizontal scaling via containerisation.	Supports up to 500 concurrent users; deployable via Docker on cloud platforms.
3	Security & Data Privacy	All API keys stored in .env files (not hardcoded). HTTPS enforced for all communications. User inputs sanitised to prevent injection attacks.	Zero hardcoded secrets; HTTPS-only; input validation on all endpoints.
4	Reliability & Availability	The application must maintain high uptime with graceful error handling when Gemini API is temporarily unavailable, returning user-friendly messages.	99% uptime target; fallback messages displayed on API errors within 5 seconds.
5	Usability & Accessibility	Interface must be intuitive, responsive across devices, and require no technical knowledge. All features accessible via simple button clicks and text input.	Usable by students aged 14+; responsive on mobile, tablet, and desktop screens.
6	Other (Maintainability)	Codebase follows modular structure with clear separation of frontend, backend, and AI layers. Comments and documentation included for each module.	Code coverage > 70%; README with setup instructions; modular file structure maintained.